

* Principle Component Analysis [PCA]

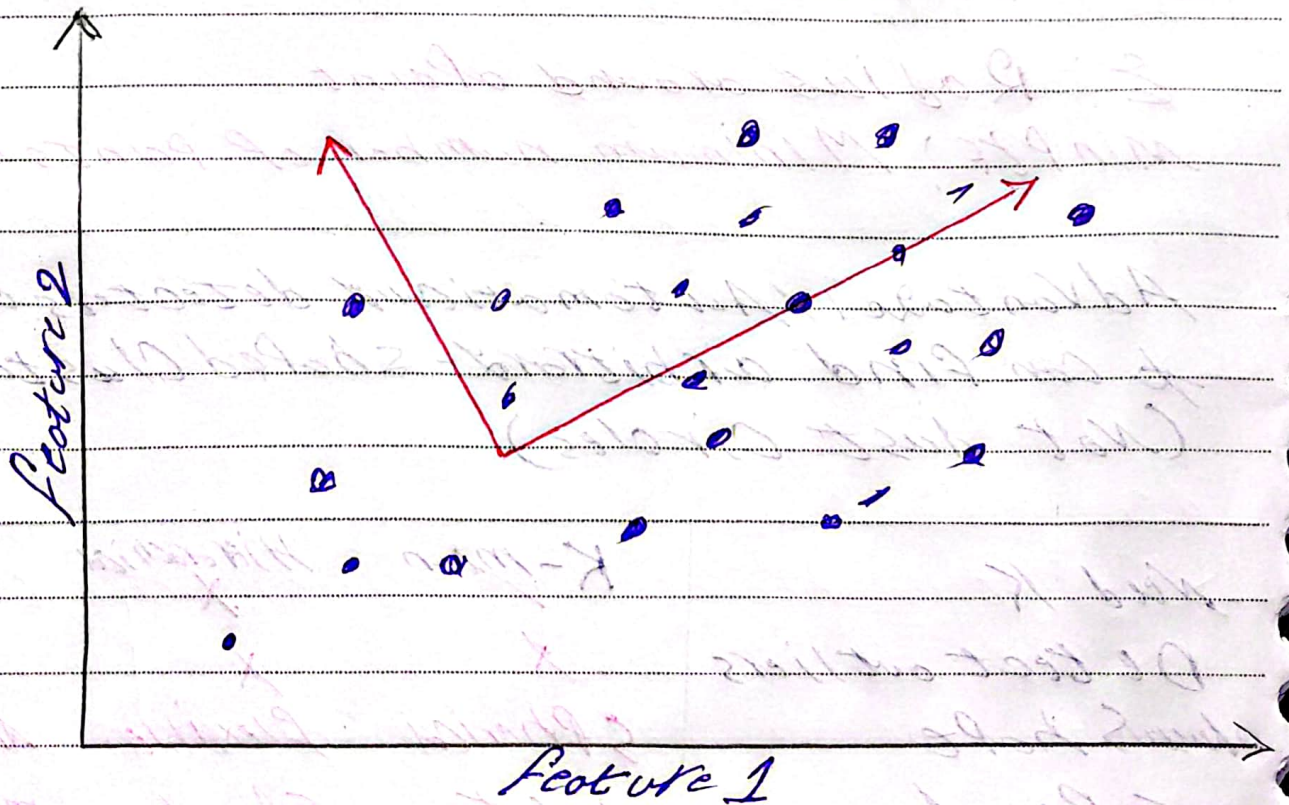
→ it is an unsupervised technique used to reduce the number of features while keeping as much information as possible

→ PCA finds new axes (direction):

* First axis (PC1) → captures maximum variance

* Second axis (PC2) → second highest variance
→ [orthogonal to PC1]

Variance = How spread out the data is



/ / التاريخ:

الموضوع:

Explained Variance:

→ Tells How much information each Component holds

$PC_1 = 70\%$, $PC_2 = 20\%$, $PC_3 = 5\%$

→ First 2 Components = 90% (usually enough)